Live Class

Number System 01

Placement & Competitive Exams

(Infosys | Wipro | TCS | Cognizant | IBPS – IT Officer | Other Placement & Govt Exams)

Updated Content: Number System



Campusmonk

Today's Roadmap

1. Quick PYQ Challenge
2. Relation b/t Dividend, Divisor
3. Factors (Total , Even , odd)
4. PYQ Infosys (Number Based Data Sufficient)
5. Divisibility Rule
6. Sum of First n Natural Numbers **(Important Formulae)**

“Question of the day” – PYQ



Question 24 :
The total numbers of square on a chessboard is:

Question: Tag – (Govt Exam/ UPSC Civil Services Examination (CSE) Prelims 2020)

One page is torn from a booklet whose pages are numbered in the usual manner starting from the first page as 1. The sum of the numbers on the remaining pages is 195. The torn page contains which of the following numbers ?

a) 5, 6

b) 7, 8

c) 9, 10

d) 11, 12

Question1: (Tag – CGL 2019 / TCS NQT 2018)

Natu and Buchku each have a certain number of oranges. Natu says to Buchku, **"If you give me 10 of your oranges, I will have twice the number of oranges left with you."** Buchku replies, **"If you give me 10 of your oranges, I will have the same number of oranges as left with you."** What is the number of oranges with Natu and Buchku, respectively?

Options:

- (A) 50, 20
- (B) 70, 50
- (C) 20, 50
- (D) 50, 70

Question 2: (CAT 1998 / SSC CHSL 2020)

A man engaged a servant on the condition that he would pay him Rs. 90 and a turban after a service of one year. He served only for nine months and received the turban and Rs. 65. The price of the turban is:

TITA (Type In The Answer)

Question 3:

In a division sum, the divisor is 10 times the quotient and 5 times the remainder. If the remainder is 48, then the dividend is:

1. 5808
2. 5809
3. 5484
4. 3472

Concept Clarity Sheet- Factor’s

Question4 :How many divisors will be there of the number 1020?(Amcat/Infosys/TCS/Govt)

(a) 12

(b) 20

(c) 24

(d) 36

Question 5:

How many divisors will be there of the number 1200?

(a) 12

(b) 30

(c) 24

(d) 36

Question 6-8:

Of the Number: N= 420

1. Find the total number of factors =
2. Find the total number of Even factors
3. Find the total number of odd factors

Concept Clarity Sheet-Integers

Question 9 : GRE/GMAT/Infosys PYQ

The positive numbers n and m are distinct prime integers.

Quantity A	Quantity B
The total number of factors of n^2m^3 that are greater than 1.	12

Question 10 : GRE/GMAT/Infosys PYQ

The positive integer R is four greater than a multiple of nine. Which of the following could be the remainder when R is divided by 6?
Indicate all such integers

(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
(F) 5

Question 11 : (Infosys SE 2025)

If x and y are positive integers, is y odd?

Statement 1: x is odd.

Statement 2: xy is odd.

A. Statement 1 alone is sufficient

B. Statement 2 alone is sufficient

C. Both statement 1 and statement 2 together are sufficient

D. Both statement 1 and statement 2 even together are not sufficient

Question 12: *(Infosys SE PYQ)*

Is ba^b + b^{b+1} is even ? (If a , b are positive integers)

I. If b is even

II. If a is even

A. Statement I alone is sufficient

B. Statement II alone is sufficient

C. Both statement I and statement II together are sufficient

D. Both statement I and statement II even together are not sufficient

E. Either I alone or II alone is sufficient to answer the question .

Question 13: CAT/Infosys

If n= 1+x, where x is the product of four consecutive positive integers, then which of the following statement is/are true

1. n is odd

2. n is prime

3.n is a perfect square

A. 1 only

B. 2 only

C. 3 only

D. 1&3 only

Concept Clarity Sheet Divisibility Rule (2, 4 ,8)

Question 14 :

A hundred digit number is formed by writing first 54 natural numbers one after the other as 123456……5354.Find the remainder when this number is divided by 8 ?

(a) 4 (b) 2

(c) 7 (d) 0

Concept Clarity Sheet Divisibility Rule (3,9)

Question 15:

What least value must be assigned to * so that the number **451*603** is exactly divisible by 9?

(a) 7 (b) 8 (c) 5 (d) 9

Question 16:

If **5432*7** is divisible by 9, then the digit is place of * is ?

(a) 0 (b) 1 (c)6 (d) 9

Question 17:

When 335 is added to 5A7 the result is 8B2.

8B2 is divisible by 3.

What is the Largest and the smallest possible value of A?

Concept Clarity Sheet Divisibility Rule (11)

Question 18:

Find the value of k in the number

9314k8025 so that the number is divisible

by 11

Question 19:

If the nine-digit number **259876P05** is completely divisible by 11, then what is the value of $(p^2 + 5)$?

A. 58

B. 56

C. 76

D. 54

Composite & Co-prime No

Question 20:

Which of the given values Is divisible by 15?

A. 2365

B. 1375

C. 4365

D. 2275

Question 21:

Which of the given values Is NOT divisible by 18?

A. 54036

B. 50436

C. 34056

D. 65043

Question 22:

Which of the following number is divisible by 24 ?

A. 35718

B. 63810

C. 537804

D. 3125736

Concept Clarity Sheet

Concept Clarity Sheet

Question 23:

The Largest number of four digits exactly divisible by 88 is ?

A. 9988

B. 9944

C. 8888

D. 9768

Question 24:

If a 10-digit number 5432y1749x is divisible by 72, then what is the value of (5x-4y)?

Question 25:

A number **N = 897324P64Q** is divisible by both 8 and 9. Which of the following is the value of P + Q?

(i). 2 (ii). 11 (iii). 9

(a) either (i) or (ii)

(b) either (ii) or (iii)

(c) either (i) or (ii) or (iii)

(d) None of these

Question 26:

How many pairs (A, B) are possible in the number 89765A4B

if the number is divisible by 9 and it is given that the last digit of the number is even?

Ans:

Question 22:

How many pairs (A, B) are possible in the number 89765A4B

if the number is divisible by 9 and it is given that the last digit of the number is even?

Ans:

Question :

If a 9-digit number 389x6378y is divisible by 72, then the value of 6x + 7y is:

A. 16

B. 32

C. 28

D. 64

E. None of these

Question :

If the number 6p5157q is divisible by 88,

then p x q= ____,

where p and q are single digit numbers.

A. 16

B. 18

C. 28

D. 64

E. None of these

Question :

If **3236A187B** is divisible by **144,** find the values of A and B respectively.

(a) (2, 4)

(c) (4, 4)

(b) (4, 2)

(d) None of these

p, q, and r are integers
p + qr is odd
pq + qr is even

Quantity A	Quantity B
$(-1)^p$	0



Question 24 :

The total numbers of square on a chessboard is:

Concept Clarity Sheet

6489

380247